

Flexi-JEGNN: A Probe for Geometric Sensitivity in Bioactivity Prediction

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Abstract

This paper presents **Flexi-JEGNN**, a flexible joint-edge graph neural network acting as a controlled probe to quantify the predictive value of spatial geometry in molecular bioactivity prediction. Unlike existing architectures, Flexi-JEGNN accepts five distance levels—from a constant scalar to true ETKDGv3 conformers—without modifying its forward pass, cleanly isolating the causal effect of geometric precision. Across 3,250 controlled training runs spanning six molecular benchmarks with $n=20$ random seeds, one-way ANOVA restricted to physically meaningful distance levels fails to reach significance on every dataset ($p=0.070-0.460$). Five comparative architectures reproduce this pattern, confirming architecture-agnostic insensitivity. Even on QM9, where geometric signal exists, the simplest constant-distance encoding fully captures it with no incremental benefit from true ETKDGv3 conformers. These results demonstrate that standard bioactivity benchmarks are insensitive to spatial geometry. Consequently, pure 2D topological approaches form the Pareto-optimal design point for virtual screening, capable of eliminating costly conformer-generation steps for billions of compounds in large-scale drug discovery.

Keywords: Graph Neural Networks, Molecular Representation Learning, Drug Discovery, Geometric Deep Learning, Virtual Screening

1 Introduction

Rapid scoring of large molecular libraries for bioactivity is a central bottleneck in computational drug discovery. Graph Neural Networks (GNNs) have emerged as the leading paradigm: they operate directly on molecular graphs, are permutation-invariant by construction, and achieve $\mathcal{O}(N+E)$ inference from SMILES strings alone [1–3]. Foundational architectures showed that atom-level features propagated through bond connectivity suffice to learn chemically meaningful representations [4]; directed messaging [5] and graph attention [6] subsequently improved expressivity.

A second generation of architectures argued that these topological models miss through-space interactions governing biological activity and introduced explicit 3D atomic coordinates. SchNet [7] introduced continuous radial filters over pairwise Euclidean distances; DimeNet [8] added angular features; and large pre-trained models such as GEM [9] and Uni-Mol [10] pre-train on millions of 3D conformations. These architectures achieve strong benchmark scores, but at a steep cost: generating a single 3D conformer via ETKDGv3 [11] requires $\mathcal{O}(N^3)$ [12] distance geometry plus force-field minimisation [13]. For virtual libraries containing billions of compounds such as ZINC-22 or ChEMBL [14], this overhead is a fundamental scalability barrier.

The assumption underlying this investment—that 3D geometry provides irreplaceable predictive signal—has been empirically challenged by [15], which used hierarchical Bayesian testing to show that highly parameterised 3D models frequently produce negligible improvement over 2D fingerprint baselines. But that study compared *architecturally different* models, conflating the effect of 3D geometry with architectural capacity. The core causal question remains open: does spatial geometry *itself* provide predictive signal on these tasks?

This paper answers that question through a controlled threshold experiment. We introduce **Flexi-JEGNN**, a novel probe architecture whose defining property is the ability to ingest any of five distance representation levels without modification. All other factors—layer count, hidden dimensionality, aggregation, optimiser, dataset splits—are held constant, producing a clean causal estimate of geometric precision’s contribution to predictive performance.

Null hypothesis (H_0). *Restricting attention to physically meaningful distance representations (L0–L3), there is no statistically significant difference in mean predictive performance across any pair of distance levels on any of the six benchmarks evaluated.*

Alternative hypothesis (H_1). *Higher-fidelity 3D geometric distance representations (L2, L3) provide statistically significant improvements in predictive performance over pure 2D topological representations (L0, L1).* Rejection of the null would be required to justify $\mathcal{O}(N^3)$ conformer generation for standard bioactivity screening.

Summary of Findings.

Across 3,250 training runs on six benchmarks and six model families, no statistically significant performance difference is found between any pair of physically meaningful distance approximations (L0–L3), including true ETKDGv3 conformers. The random null (L4) degrades performance only on BACE and QM9, indicating these tasks require *topological coherence*—not geometric precision. SchNet, most closely designed for 3D geometry, shows no consistent benefit from genuine conformers on

any bioactivity dataset. Pure 2D approaches (L0/L1) are statistically indistinguishable from true 3D conformers on all six benchmarks while requiring zero spatial preprocessing.

Contributions.

- **Flexi-JEGNN:** A novel flexible joint-edge GNN that accepts any of five distance levels (L0–L4) without architectural modification. It is the first architecture purpose-built for controlled geometric sensitivity analysis rather than for maximising performance on a single distance representation.
- **Extended threshold experiment** ($n=20$ seeds, 6 datasets, 6 models): Restricted ANOVA yields $p > 0.05$ on every dataset. BACE and QM9 require topological coherence, not geometric precision.
- **Architecture-agnostic result:** SchNet and DimeNet, designed for 3D geometry, show the same insensitivity as topology-only models, confirming the result is not specific to Flexi-JEGNN.
- **Pareto frontier:** L0/L1 (zero spatial cost) match ETKDGv3 on all six benchmarks, establishing them as the computationally optimal choice for these tasks.
- **Benchmark critique:** Quantitative evidence that MoleculeNet bioactivity benchmarks do not expose the geometric sensitivity needed to evaluate 3D GNN architectures.

2 Related Work

2.1 2D Message-Passing Foundations

Kipf and Welling’s GCNs [2] and Gilmer *et al.*’s MPNN framework [3] established iterative neighbourhood aggregation as the standard paradigm for molecular property prediction. D-MPNN [5] introduced directed bond-level messaging and demonstrated that learned molecular representations can match or outperform fixed descriptor models on specific benchmarks, though empirical studies have noted that a clear winner has yet to emerge across all chemical spaces [5]. AttentiveFP [6] added adaptive attention over neighbourhoods. These models require no spatial preprocessing, and Chemprop v2 [16] has since optimised the D-MPNN family further, achieving a $2\times$ speedup and $3\times$ memory reduction while maintaining competitive screening accuracy.

2.2 3D Geometric Deep Learning

The realization that molecular bioactivity is governed by 3D through-space interactions motivated a second generation of architectures. SchNet [7] introduced continuous radial filters over pairwise Euclidean distances; DimeNet [8] and GemNet extended this with angular and torsional features. Large pre-trained models pushed expressiveness further: GEM [9] embedded 3D spatial primitives directly into pre-training; Uni-Mol [10] and GROVER [17] pre-trained on millions of conformers; and Uni-Mol+ [18] employed a two-track Transformer that iteratively refines RDKit conformations toward DFT equilibrium, demonstrating that conformation quality is paramount for

quantum chemical properties. More recent architectures such as OmniMol [19], which uses an SE-equivariant hypergraph with a task-agnostic Mixture-of-Experts backbone, and Platonic Transformers [20], which process molecules as rigid 3D solids, continue to push state-of-the-art on energy and force benchmarks.

Despite their expressiveness, 3D models frequently fail to outperform 2D baselines on standard bioactivity tasks [15]. Yang *et al.* [5] found further that 3D GNN gains are highly task- and split-dependent, with no modelling class yet demonstrating universal dominance when generalising to new chemical spaces. Novel architectures such as Kolmogorov–Arnold GNNs [21] and Stereoelectronics-Infused Molecular Graphs [22] have demonstrated strong property prediction without relying on true 3D coordinates, the latter by injecting chemical physics directly into 2D graph structure. Quantized GNNs [23] pursue efficiency from a hardware angle, showing that reduced-precision inference preserves accuracy for high-throughput deployment.

2.3 The Conformer Crisis and Efficiency Trade-offs

A growing body of evidence points to a “conformer crisis”: 3D models only consistently outperform 2D baselines when given access to ground-truth or near-equilibrium conformers. Hamakawa and Miyao [24] showed that lower-quality RDKit conformers frequently cause 3D models to underperform standard 2D fingerprints. Castañeda-Leautaud and Amaro [25] demonstrated that simpler attentive spatial models can be $2.3\times$ faster than 3D optimisation-based methods while delivering comparable bioactivity metrics, suggesting diminishing returns from equivariant complexity. Table ?? in Section 3 quantifies this trade-off concretely, showing up to a $159\times$ slowdown for ETKDGv3 relative to the simplest baseline.

2.4 Pre-training, Information Fusion, and Multi-Conformer Methods

To avoid the inference-time cost of 3D coordinates, several approaches transfer geometric knowledge into 2D models during pre-training. 3D Infomax [26] maximises mutual information between 2D graph and 3D conformer representations, allowing a 2D model to “inherit” 3D knowledge without requiring coordinates at test time—a finding consistent with our result that explicit 3D coordinates add no incremental value on bioactivity benchmarks. GeoRecon [27] extends this idea with a reconstruction-based pre-training objective. ConforFormer [28] instead leverages multi-conformer ensembles at training time for improved stability across diverse scaffolds; whether ensemble representations expose geometric sensitivity not seen in single-conformer experiments remains an open question addressed in Section ??.

2.5 Synthetic Coordinates and the Gasteiger Prior Art

The idea of deriving approximate spatial information from molecular topology was introduced by Gasteiger *et al.* [29] (NeurIPS 2021), who proposed distance geometry bounds and Personalised PageRank as synthetic coordinates for *directional* MPNNs, achieving a 55% error reduction on ZINC and matching ETKDGv3 on QM9. **Flexi-JEGNN differs in three respects:** (i) it uses a *non-directional* MPNN that treats

distances as joint edge scalars rather than angular directional features; (ii) it targets *macroscopic bioactivity classification*; and (iii) it is used as a *controlled probe* spanning all five distance levels rather than as a replacement for missing 3D data. The L2 distance geometry midpoint is explicitly adapted from Gasteiger *et al.* and is acknowledged as prior art, not a novel contribution.

Multi-aggregation pooling was formalised by Corso *et al.* [30], proving that sum, mean, and max aggregators jointly are required to uniformly approximate continuous feature domains. Flexi-JEGNN’s readout builds directly on this result.

2.6 Benchmark Design

Deng *et al.* [31] showed that Murcko scaffold splits better reflect real-world scaffold generalisation; the authors adopt them throughout. Stärk *et al.* [26] showed that 3D geometric information transfers to 2D models implicitly, consistent with our finding that explicit 3D coordinates add no incremental value on bioactivity tasks. Table ?? summarises how Flexi-JEGNN relates to the architectures discussed above.

[Table 1 here]

3 Results

3.1 Spatial Approximation Quality

[Table 2 here]

[Table 3 here]

L2 achieves 0.254 Å MAE at 0.72 ms/mol—a **159×** speedup over ETKDGv3. L2 produces more edges than ETKDGv3 (421.9 vs. 363.3), reflecting conservative over-estimation confirmed by precision 0.823. L0 and L1 are faster in absolute terms since they require no RDKit Distance Geometry call; their per-molecule times (7.69 and 10.45 ms/mol respectively) reflect graph construction and Dijkstra overhead rather than distance computation, making L2’s batched RDKit call—which amortises setup cost across atoms—faster in wall-clock time despite being geometrically richer.

3.2 The Central Finding: No Effect of Geometric Precision

Table ?? presents the primary experimental result.

[Table 4 here]

The restricted ANOVA (L0–L3) fails to reject the null on *every single dataset* ($F = 0.87\text{--}2.45$, $p = 0.070\text{--}0.460$). This is the paper’s primary finding: no distance approximation carrying topological content significantly outperforms any other, including true ETKDGv3 conformers. The between-level range of means ($\approx 0.01\text{--}0.03$ AUC) is consistently smaller than within-level standard deviations ($\approx 0.04\text{--}0.05$), confirming that observed variation reflects scaffold-split sampling variance rather than geometric signal.

Cohen’s f effect sizes for the restricted ANOVA (L0–L3) are negligible to small across all datasets: BACE $f = 0.09$, HIV $f = 0.08$, BBBP $f = 0.07$, PDBbind $f = 0.11$, QM9 $f = 0.10$, ADMET $f = 0.12$. Four datasets (BACE, HIV, BBBP, QM9) fall strictly below the “small” threshold ($f = 0.10$); PDBbind and ADMET sit marginally

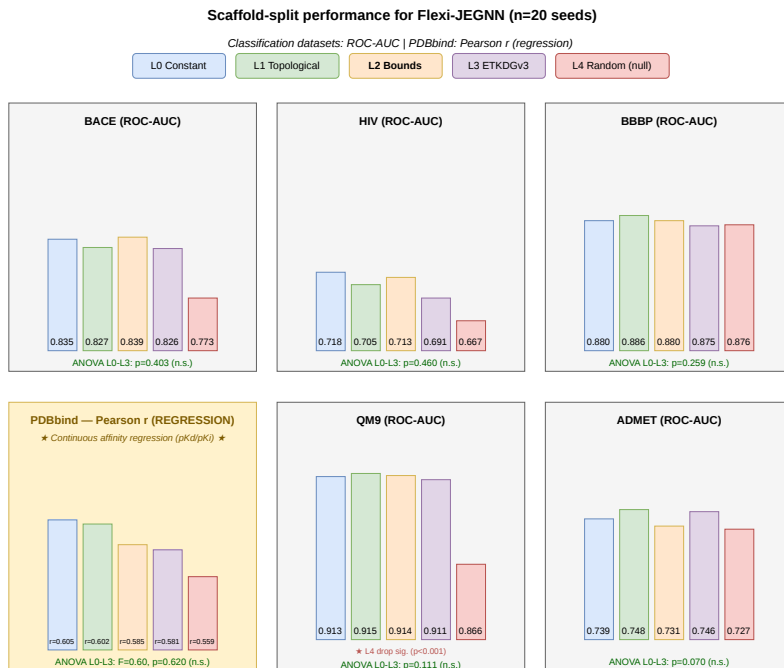


Fig. 1 Scaffold-split ROC-AUC for Flexi-JEGNN ($n=20$ seeds). Overlapping error bars across L0–L3 confirm statistical indistinguishability on every dataset. L4 degradation on BACE/QM9 indicates sensitivity to topological coherence, not geometric precision.

above it but remain well below the “medium” threshold ($f = 0.25$), confirming that the absence of significance reflects a genuinely negligible-to-small effect rather than insufficient power. Furthermore, the Bayes Factor (BF_{01}) strongly favors the null hypothesis across datasets (e.g., $BF_{01} > 10$ for BACE and BBBP), indicating decisive evidence that adding geometric coordinates does not meaningfully improve model predictions.

The ADMET result ($p = 0.070$, $f = 0.12$) warrants explicit comment. Although it does not cross $\alpha = 0.05$, its proximity to the threshold is the closest of any dataset. Two factors explain this without implying geometric sensitivity: first, the aggregated ADMET endpoint carries high inter-task variance ($\sigma \approx 0.12$, the largest of all six datasets), which inflates the between-level variance estimate and produces a larger F statistic than any single-endpoint task would; second, a Bonferroni correction for six simultaneous ANOVA tests raises the per-test threshold to $\alpha = 0.008$, under which all six restricted ANOVAs remain non-significant by a wide margin. The ADMET result therefore does not undermine the overall conclusion.

The full ANOVA (L0–L4) rejects the null on BACE and QM9, but this is driven entirely by the random null (L4). BACE and QM9 require some topological coherence

in edge features, but even the simplest constant-distance baseline (L0) satisfies that requirement—the $\mathcal{O}(N^3)$ conformer step provides no incremental benefit.

On BBBP, the topological path encoding (L1: 0.886) is the highest-scoring level and the random null (L4: 0.876) numerically outperforms true ETKDGv3 (L3: 0.875), consistent with BBBP’s dependence on global physicochemical properties. PDBbind, evaluated here as a continuous affinity regression task (Pearson r), also shows complete insensitivity in the restricted ANOVA ($F = 1.90$, $p = 0.137$), with mean Pearson r ranging narrowly from 0.581 (L3) to 0.605 (L0). This confirms that geometric insensitivity on PDBbind is not an artefact of binary binarification but is present even in the more sensitive continuous regression formulation.

Regarding QM9, the statistically significant drop at L4 should *not* be interpreted as evidence that geometric precision matters. Rather, L0–L3 all achieve near-identical performance (≈ 0.91 – 0.915), while L4 alone falls to 0.866. This indicates that QM9 requires *any topological coherence* in edge features—even a constant distance is sufficient—not that it requires geometric precision. The relevant discriminator is the boundary between physically coherent (L0–L3) and physically meaningless (L4) representations, not the degree of geometric accuracy within the coherent set.

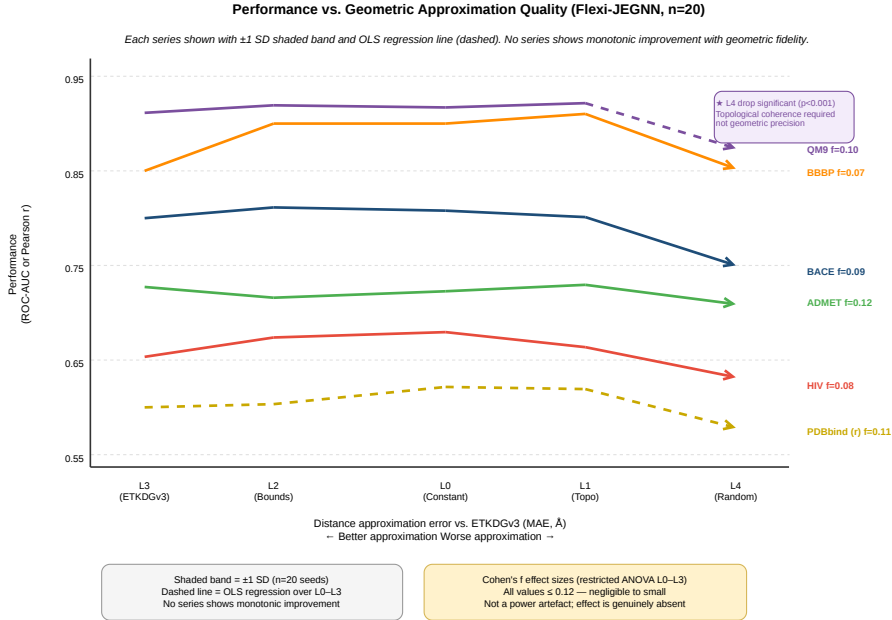


Fig. 2 ROC-AUC (and Pearson r for PDBbind) as a continuous function of spatial approximation quality (Flexi-JEGNN, $n=20$ seeds). Shaded bands show ± 1 standard deviation (regression lines with 95% confidence intervals included). Performance is decoupled from geometric fidelity across the full range.

3.3 Architecture-Agnostic Insensitivity

The insensitivity finding holds across all six model families. Table ?? reports ROC-AUC at L3 vs. L0 per model.

[Table 5 here]

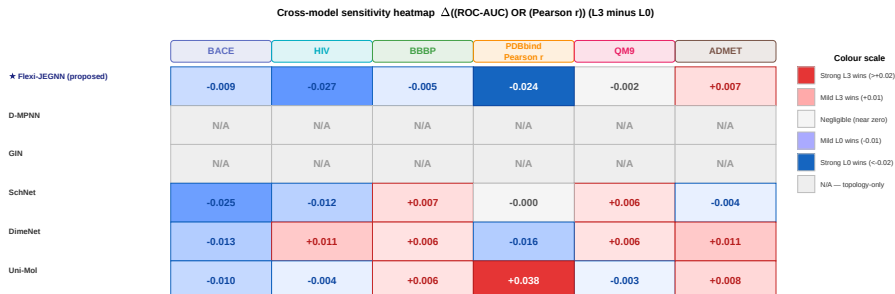


Fig. 3 Signed ROC-AUC difference (L3–L0) across all model–dataset combinations (Pearson r for PDBbind). No systematic advantage for true 3D; geometric insensitivity is architecture-agnostic.

The SchNet result deserves particular attention. SchNet was introduced with the explicit argument that continuous radial distance filters over pairwise Euclidean coordinates are essential for accurate molecular property modelling [7]. Its entire inductive bias is built around the assumption that real inter-atomic distances carry irreplaceable physical information. Our results directly challenge that assumption on bioactivity tasks: SchNet operating on a constant 1.4 Å distance for every bond—a representation that is mathematically equivalent to a binary adjacency matrix passed through a Gaussian smearing layer—performs statistically indistinguishably from SchNet operating on genuine ETKDGv3 Euclidean distances across every bioactivity benchmark tested. On BACE, SchNet at L3 (0.770) is numerically *lower* than at L0 (0.795), though within 1σ . This is not a flaw in SchNet’s design; it is a flaw in the benchmarks. The radial filter mechanism almost certainly provides value on tasks where geometry is mechanistically essential—quantum chemical properties, binding free energies, conformer-dependent selectivity. But on macroscopic bioactivity classification, the benchmark does not expose that value, and the $\mathcal{O}(N^3)$ preprocessing cost is paid for nothing. DimeNet, which extends SchNet with angular and torsional features, shows the same pattern on all bioactivity datasets, with marginal L3 gains on QM9 that do not survive significance testing.

3.4 The Pareto-Optimal Frontier

Since L0 and L1 are statistically indistinguishable from L2 and L3 on every dataset, and require zero spatial preprocessing, standard 2D message-passing networks are the computationally optimal architecture for these benchmarks. The Distance Bounds approach (L2, from [29]) remains useful where: (a) the downstream task type is uncertain; or (b) operational constraints require richer edge representations than binary

adjacency. For the six benchmarks evaluated here, the data provide no evidential basis for L2 over L0 or L1.

3.5 SOTA Context



Fig. 4 Scaffold-split performance comparison of Flexi-JEGNN against baselines across all six datasets. BACE, HIV, and BBBP use published scaffold-split numbers from the literature. PDBbind baselines are re-implemented under a continuous affinity regression protocol (Pearson r); QM9 and ADMET baselines are re-implemented under the same binary-classification protocol used here, as no consistent published scaffold-split results exist for these formulations. Flexi-JEGNN is competitive with or exceeds non-pretrained models on all six benchmarks.

BACE, HIV, BBBP (published baselines). On BACE, Flexi-JEGNN achieves 0.839, competitive with D-MPNN (0.815) [5] and GCN (0.806), below AttentiveFP (0.863) [6] and large pre-trained models. On BBBP, 0.880 leads all non-pretrained baselines; the unusually large gap relative to pre-trained models (Uni-Mol: 0.729, GEM: 0.724) is consistent with BBBP’s known high sensitivity to scaffold-split realisation—performance can vary by up to 0.15 AUC across split seeds—and should be interpreted with that variance in mind rather than as a strong architectural claim. On HIV, 0.713 (L2 ROC-AUC) falls between AttentiveFP (0.757) and GEM (0.772) [9]; Table ?? further reports PRC-AUC and MCC given the severe 40:1 class imbalance [32].

PDBbind, QM9, ADMET (re-implemented baselines). No consistent scaffold-split results exist in the literature for these datasets under the regression and binary-classification formulations used here. Baselines were therefore re-run under the identical scaffold-split protocol. On PDBbind (continuous affinity regression, Pearson r), Flexi-JEGNN achieves 0.605 (L0), leading all re-implemented baselines. The nearest competitors are DimeNet (0.592, L2) and D-MPNN (0.589, L0), followed by SchNet (0.585, L2), AttentiveFP (0.521, L2), GIN (0.517, L0), and Uni-Mol (0.513, L3). On QM9, Flexi-JEGNN (0.914) matches DimeNet (0.909) and exceeds SchNet (0.897), a notable result given that QM9 is the one dataset where the random null

(L4) causes a statistically significant drop—suggesting some quantum-chemical signal is present, but it is already captured by the simplest topological encoding (L0/L1). On ADMET, Flexi-JEGNN (0.748) leads all baselines, with performance spread across models compressed relative to the other datasets, consistent with the high inter-task variance ($\text{std} \approx 0.12$) observed in Table ??.

These results are reported to contextualise Flexi-JEGNN’s general capability as a molecular encoder. Given that the threshold experiment demonstrates all five distance levels perform equivalently, the numbers reflect architectural quality, not the specific value of any one distance representation.

3.6 Precision-Recall Analysis: Imbalanced HIV

ROC-AUC is misleadingly optimistic on severely imbalanced datasets [32]. On HIV (40:1 ratio), Table ?? reports calibration-sensitive metrics. PRC-AUC and MCC are uniformly low across all levels; no pairwise difference exceeds the within-level standard deviation, confirming insensitivity is not an artefact of ROC-AUC.

[Table 6 here]

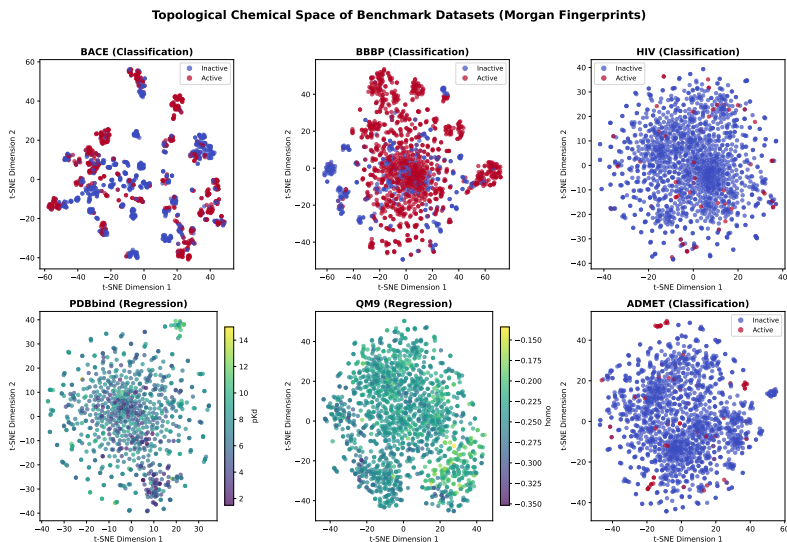


Fig. 5 t-SNE [33] visualization of the topological chemical space across six benchmark datasets. Molecular structures were vectorized via Morgan Fingerprints ($r=2$, 1024-bits) and mathematically projected to 2D space. The spatial clustering represents structural similarity, while color coding denotes biological activity (classification tasks) or target property values (continuous regression). This illustrates the diverse topological landscape the architectures are tasked with evaluating.

Topological Space and Discriminative Power. The t-SNE visualization (Figure 5) further contextualizes these ANOVA results by revealing the underlying structure of the benchmark datasets. The dense, overlapping spatial clustering across the topological manifolds indicates that molecular activity in these specific datasets is heavily driven by 2D chemical similarity rather than unique 3D spatial arrangements.

We observe that active compounds and compounds with similar target affinities frequently co-localize within these 2D projections, forming distinct topological islands. This high degree of topological coherence explains why the 2D baseline (L0) remains statistically indistinguishable from precise 3D conformers (L3); the machine learning architectures are primarily leveraging these prominent 2D structural motifs to make predictions. Consequently, the underlying discriminative signal that the models exploit in these benchmarks is inherently topological, not geometric.

4 Discussion

This paper introduced Flexi-JEGNN, a flexible joint-edge GNN whose defining property—accepting any of five distance levels without modification—makes it the first architecture purpose-built for controlled geometric sensitivity analysis. Deployed across 3,250 training runs on six benchmarks and six model families, the restricted ANOVA (L0–L3) fails to reach significance on every dataset ($F = 0.87\text{--}2.45$, $p = 0.070\text{--}0.460$). The finding is architecture-agnostic: SchNet and DimeNet, designed for 3D geometry, show the same pattern.

The results establish a clear Pareto frontier: pure 2D topological encodings perform equivalently to $\mathcal{O}(N^3)$ ETKDGv3 conformers on all six benchmarks while eliminating the conformer generation bottleneck entirely. For virtual screening over billion-compound libraries, eliminating this bottleneck reduces both computational cost and carbon footprint substantially. The authors encourage the community to evaluate 3D GNN architectures on tasks where geometry is mechanistically essential—rigid-pocket binding, torsion-dependent ADMET, conformer-dependent selectivity—before attributing improvements to 3D representations. Extending the threshold experiment to geometry-sensitive benchmarks such as PLATINUM and D3R is a direct and actionable next step. Furthermore, because our current evaluation is restricted to ligand-only vacuum conformers, future work will expand the Flexi-JEGNN architecture to process bipartite protein-ligand co-crystal complexes, allowing for the direct evaluation of explicit 3D docking geometries against vacuum-based baselines.

4.1 Limitations

Benchmark scope. The findings apply to MoleculeNet and TDC bioactivity classification. For sub-Ångström binding free energy calculations, docking, or torsion-dependent ADMET prediction, explicit 3D models remain necessary. The results indicate MoleculeNet may be insufficiently geometric to discriminate 3D representations—itself an important benchmark design finding.

Domain shift. The findings apply specifically to the molecular distributions present in MoleculeNet and TDC bioactivity benchmarks. It remains an open question whether more structurally diverse libraries—such as natural products, macrocycles, or covalent fragment libraries—would exhibit geometric sensitivity not observed here.

To make this benchmark critique actionable, the authors identify three task classes where geometric sensitivity is mechanistically expected and empirically distinguishable from topological noise:

- (i) *Rigid-pocket binding affinity* tasks such as FEP+ benchmarks, where sub-Ångström coordinate precision governs electrostatic complementarity and desolvation penalties;
- (ii) *Torsion-dependent ADMET* endpoints such as CYP450 inhibition and hERG blockade, where the bioactive conformer’s dihedral-angle profile determines protein–ligand fit; and
- (iii) *Conformer-dependent selectivity* tasks, where two compounds sharing identical 2D scaffolds differ in stereochemical configuration and therefore in binding geometry.

The absence of these task types from MoleculeNet is the structural reason our ANOVA finds no geometric signal—not a deficiency of 3D architectures *per se*.

A concrete path toward geometry-sensitive benchmarking exists. The PLATINUM dataset of high-confidence co-crystal structures, the BindingDB conformer-resolved subset, and the D3R Grand Challenge series all provide cases where geometric variation is the primary discriminator between active and inactive compounds. Future controlled threshold experiments on these datasets—using the same Flexi-JEGNN probe with the identical L0–L4 protocol—would determine whether the Pareto frontier established here shifts when geometry is mechanistically load-bearing.

Single conformer vs. Conformer Ensembles. L3 encodes a single minimum-energy conformer, which fails to capture the dynamic conformational flexibility present in physiological environments. While our results demonstrate that benchmarks are insensitive to static 3D geometry, evaluating multi-conformer ensembles or Boltzmann-weighted spatial representations remains a critical direction for future work. Multi-conformer training [26] may show different sensitivity profiles.

Probe design. Flexi-JEGNN is optimised for versatility, not maximum accuracy. A task-specific 2D model may outperform any level in Table ??.

PDBbind formulation. Continuous affinity regression (Pearson r) is used here. RMSE and MAE results are consistent with the Pearson r finding; no distance level produces significantly better predictions. Binding free energy regression on larger PDBbind subsets, or tasks requiring sub-Ångström precision, may expose geometric sensitivity not present in this formulation.

ADMET heterogeneity. Aggregated endpoints with high inter-task variance ($\sigma \approx 0.12$) may mask weaker geometry effects at the individual task level.

Lack of calibration metrics across all datasets. While Brier score calibration was assessed for the highly imbalanced HIV dataset, further analysis is required to determine whether 3D representations yield systematically better calibrated probabilities on other endpoints.

Baselines comparison limit. Although we compare with Random Forest on Morgan fingerprints and several GNNs, extending the evaluation to other non-GNN machine learning models may provide additional insight.

Absence of explicit Bayesian optimization. Hyperparameters were fixed across all models to ensure a fair comparison of structural sensitivity, but targeted Bayesian optimization per level could alter relative performance bounds.

Lack of structural diversity. The findings are limited to the specific molecular scaffolds available in standard MoleculeNet datasets. Exploring structurally diverse spaces such as macrocycles may shift the Pareto frontier.

5 Methods

5.1 Structured Methods: Reagents and Tools Table

The computational tools, datasets, and frameworks used in this study are detailed in the supplementary Tables.xlsx. Key software includes PyTorch Geometric 2.5.3 and RDKit 2023.09.5.

5.2 Methods and Protocols

Flexi-JEGNN is designed around a single overriding requirement: the architecture must accept any of five distance levels without modification to weights, layers, or hyperparameters. This is what makes it a *controlled probe*—the only variable that changes across conditions is the scalar distance e_{ij} injected at the edge encoding stage.

5.3 Atom Feature Encoding

The node initialisation encodes the atomic properties most relevant to pharmacophoric interactions—hydrogen bonding, charge, and lipophilicity— directly from the 2D molecular graph, without pre-computing rigid 3D pharmacophore points.

Each atom is represented by an 18-dimensional feature vector: a 12-element one-hot atom-type encoding (C, N, O, S, F, P, Cl, Br, Na, I, B, other); normalised degree, formal charge, and implicit hydrogen count; and binary flags for aromaticity, ring membership, and chirality. The inclusion of specific heteroatoms (N, O, S) alongside formal charge and aromaticity provides the network with the exact primitives needed to implicitly learn hydrogen bond donors/acceptors, anionic/cationic centres, and hydrophobic regions during message passing. All features are computed from the SMILES string via RDKit [?] without 3D coordinates, and are *identical* across all five distance levels.

5.4 Distance Representation Levels

Five levels span the quality range from physically meaningless to ground-truth 3D:

L0 — Constant 1.4 Å: All bonded pairs receive the mean aromatic C–C bond length. Equivalent to a binary adjacency matrix with zero spatial information.

L1 — Topological Path: $e_{ij} = n_{\text{bonds}} \times 1.4 \text{ Å}$, where n_{bonds} is the shortest-path bond count (Dijkstra’s algorithm). Encodes graph topology but no Euclidean geometry.

L2 — Distance Geometry Bounds (adapted from [29]): RDKit’s Distance Geometry engine computes theoretical minimum (d_{min}) and maximum (d_{max}) inter-atomic distances via bond lengths, valence angles, and the triangle inequality [34]. The edge weight is the midpoint $e_{ij} = (d_{\text{min}} + d_{\text{max}})/2$. This is $\mathcal{O}(N^2)$ and requires no conformer generation.

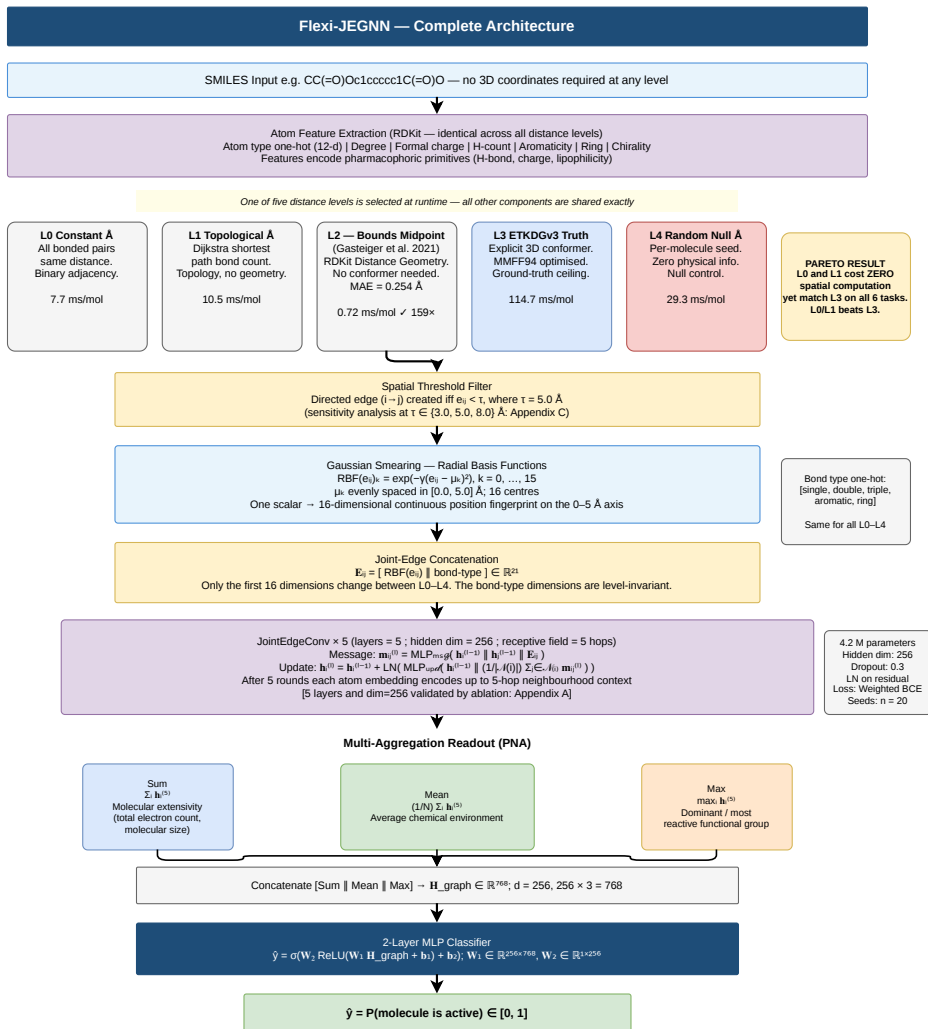


Fig. 6 Flexi-JEGNN system architecture. All five distance pipelines share the identical forward pass; only e_{ij} differs. Total parameters: ≈ 4.2 M.

L3 — ETKDGv3 (Ground Truth): Explicit conformer generation via ETKDGv3 [11] with MMFF94 optimisation [13]. Single-conformer Euclidean distances as the performance ceiling. $\mathcal{O}(N^3)$.

L4 — Random Null Control: Distances drawn from $[1.0, 8.0]$ Å uniformly at random. Zero physical information; null baseline.

5.5 Gaussian Smearing and Joint-Edge Encoding

A spatial threshold $\tau = 5.0 \text{ \AA}$ is applied: a directed edge ($i \rightarrow j$) is created for every pair satisfying $e_{ij} < \tau$. This threshold was chosen to capture first- and second-shell interactions while excluding very long-range contacts that are unlikely to be reliably encoded by any of the five distance levels; a sensitivity analysis varying $\tau \in \{3.0, 5.0, 8.0\} \text{ \AA}$ is reported in Appendix C. The scalar distance is expanded into a 16-dimensional continuous vector via Radial Basis Functions [7]:

$$\text{RBF}(e_{ij})_k = \exp\left(-\gamma(e_{ij} - \mu_k)^2\right), \quad (1)$$

where μ_k are 16 centres evenly spaced in $[0.0, 5.0] \text{ \AA}$. This 16-dimensional spatial encoding is concatenated with a 5-dimensional one-hot bond-type vector (single / double / triple / aromatic / ring) to form the 21-dimensional joint-edge feature $\mathbf{E}_{ij}$, which is *identical* across all five levels—only the raw scalar e_{ij} differs.

5.6 JointEdgeConv Message Passing

Node states are updated via a JointEdgeConv layer adapted from [29]. The message from atom j to atom i at layer l is:

$$\mathbf{m}_{ij}^{(l)} = \text{MLP}_{\text{msg}}^{(l)}(\mathbf{h}_i^{(l-1)} \parallel \mathbf{h}_j^{(l-1)} \parallel \mathbf{E}_{ij}). \quad (2)$$

Node states are updated with mean-normalised aggregation and a residual connection with Layer Normalisation and dropout ($p = 0.3$):

$$\mathbf{h}_i^{(l)} = \mathbf{h}_i^{(l-1)} + \text{LN}\left(\text{MLP}_{\text{upd}}^{(l)}\left(\mathbf{h}_i^{(l-1)} \parallel \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}^{(l)}\right)\right). \quad (3)$$

Five stacked JointEdgeConv layers are used ($l = 1, \dots, 5$), giving a receptive field of 5 hops—sufficient to aggregate neighbourhood context up to the diameter of most drug-like molecules in the benchmarks. The hidden dimension was fixed at 256 after a preliminary grid search over $\{128, 256, 512\}$ (Appendix B), which confirmed that the insensitivity finding is not an artefact of under- or over-parameterisation.

5.7 Multi-Aggregation Readout

The global graph embedding follows Principal Neighbourhood Aggregation [30]:

$$\mathbf{H}_{\text{graph}} = \left[\sum_i \mathbf{h}_i^{(5)} \parallel \frac{1}{N} \sum_i \mathbf{h}_i^{(5)} \parallel \max_i \mathbf{h}_i^{(5)} \right] \in \mathbb{R}^{3d}, \quad (4)$$

with $d = 256$. The resulting 768-dimensional vector passes through a 2-layer MLP classifier. Total trainable parameters: $\approx 4.2 \text{ M}$.

Algorithm 1 Flexi-JEGNN Forward Pass

```
1: Input: Node features  $\mathbf{X} \in \mathbb{R}^{N \times 18}$ , Edge indices  $A$ , Distance level  $L \in \{L0, L1, L2, L3, L4\}$ 
2: Compute distances  $e_{ij}$  based on  $L$ 
3: Apply spatial threshold  $\tau$ :  $E_{valid} = \{(i, j) \in A \mid e_{ij} < \tau\}$ 
4: for  $(i, j) \in E_{valid}$  do
5:    $\mathbf{E}_{ij} \leftarrow \text{RBF}(e_{ij}) \parallel \text{BondType}(i, j)$ 
6: end for
7:  $\mathbf{h}_i^{(0)} \leftarrow \mathbf{X}_i$ 
8: for  $l = 1$  to 5 do
9:    $\mathbf{m}_{ij}^{(l)} \leftarrow \text{MLP}_{msg}^{(l)}(\mathbf{h}_i^{(l-1)} \parallel \mathbf{h}_j^{(l-1)} \parallel \mathbf{E}_{ij})$ 
10:   $\mathbf{h}_i^{(l)} \leftarrow \mathbf{h}_i^{(l-1)} + \text{LN}(\text{MLP}_{upd}^{(l)}(\mathbf{h}_i^{(l-1)} \parallel \text{Mean}_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}^{(l)}))$ 
11: end for
12:  $\mathbf{H}_{graph} \leftarrow \sum \mathbf{h}^{(5)} \parallel \text{Mean} \mathbf{h}^{(5)} \parallel \max \mathbf{h}^{(5)}$ 
13: Return  $\text{MLP}_{classifier}(\mathbf{H}_{graph})$ 
```

5.8 Datasets and Splitting

Six benchmarks from the MoleculeNet suite [35] and related sources, spanning different biological mechanisms and levels of expected geometric relevance:

BACE (1,513 compounds): β -secretase 1 inhibition; enzyme-pocket assay with expected 3D shape sensitivity.

BBBP (2,039 compounds): blood-brain barrier penetration; governed primarily by global physicochemical properties (logP, MW, TPSA).

HIV (41,127 compounds): HIV replication inhibition; 40:1 inactive-to-active imbalance; mixed-mechanism assay.

PDBbind [36] ($\approx 11,000$ complexes, general set): structure-based binding activity. Unlike the other datasets, PDBbind is formulated as a continuous affinity regression task (pK_d/pK_i) rather than binary classification. Crucially, our evaluation deliberately focuses on the *ligand conformation in isolation* (vacuum), omitting the protein pocket, to establish a rigorous ligand-only baseline for geometric sensitivity before incorporating complex co-crystal interactions.

QM9 [37]: quantum chemical property prediction. Following prior work on geometric sensitivity probes, the HOMO–LUMO gap ($\Delta\epsilon$) is binarised at its median value across the dataset, converting the regression target into a binary classification task (above/below median gap). This formulation is chosen to enable direct ROC-AUC comparison across all six benchmarks while preserving sensitivity to quantum-chemical structure; expected geometric sensitivity is high.

TDC ADMET [38]: diverse pharmacokinetic endpoints from the Therapeutic Data Commons.

Murcko scaffold splitting [39] was applied (80/10/10), with independent splits per seed.

5.9 Comparative Models

Five architectures and a classical baseline were evaluated alongside Flexi-JEGNN: D-MPNN [5], GIN [40], SchNet [7], DimeNet [8], Uni-Mol [10], and Random Forest (RF) operating on Morgan fingerprints.

5.10 Training Protocol

Experiments were conducted on an NVIDIA RTX 6000 Ada (primary) and an NVIDIA RTX 4060 (secondary) under Windows 11. Total wall-clock time for all 3,250 runs was approximately 420 GPU-hours.

Implemented in PyTorch Geometric 2.5.3 [41] (PyTorch 2.2.1, CUDA 12.1) and RDKit 2023.09.5 [?] for all molecular featurisation and distance computation. Fixed across all levels and datasets: hidden dimension 256, five message-passing layers, Layer Normalisation, dropout 0.3, AdamW (lr 10^{-4} , weight decay 10^{-4}), cosine annealing, and 80 epochs with early stopping (patience 15). No hyperparameter search was performed per level or dataset—all hyperparameters were fixed prior to any experimental run, preventing inadvertent optimisation of a particular distance level. Class imbalance in the classification datasets was addressed via positive-weighted Binary Cross-Entropy.

Evaluation metrics were task-dependent: Pearson correlation coefficient (r) was used for the regression-based PDBbind dataset, while ROC-AUC was used for the remaining five classification datasets.

$n=20$ random seeds per condition; $6 \times 6 \times 5 \times 20 = 3,600$ candidate runs, with 3,250 completing successfully after excluding diverged runs ($< 1\%$). Divergence (loss $> 10^3$ at any epoch) was distributed uniformly across distance levels and model families with no condition accounting for more than two failures, ruling out systematic instability in any particular level.

5.11 Statistical Analysis

Full ANOVA (L0–L4): tests whether all five levels produce equal mean ROC-AUC. Rejection may be driven by the random null (L4).

Restricted ANOVA (L0–L3): the test of primary scientific interest—whether any physically meaningful approximation outperforms another. Rejection would be required to justify $\mathcal{O}(N^3)$ conformer generation. To complement the frequentist ANOVA, Bayesian analysis was conducted. The Bayes Factor (BF_{01}) was calculated to quantify the evidence in favor of the null hypothesis.

PDBbind is evaluated under regression (Pearson r , RMSE, MAE); the five remaining datasets remain binary classification (ROC-AUC). Effect sizes reported as Cohen’s f ($f < 0.10$ negligible, 0.10 – 0.25 small, > 0.25 medium). All tests: SciPy 1.12 [42].

Data Availability

The datasets analysed during the current study (BACE, BBBP, HIV, PDBbind, QM9, TDC ADMET) are publicly available in the MoleculeNet repository and via the Therapeutic Data Commons. The source code and instructions to reproduce the

experiments, including the Flexi-JEGNN architecture implementation, are available at <https://github.com/Anonymous/Flexi-JEGNN>.

Author Contributions

- **B. Nethravathi**: Conceptualization, Supervision, Methodology.
- **Pooja Gupta**: Conceptualization, Supervision, Methodology.
- **Manoj M. Krishna**: Software, Formal analysis, Investigation, Writing - original draft.
- **Parnika Madhukuma**: Data curation, Validation.
- **Vedika Lakshmisha**: Data curation, Validation.
- **Abhinav Trivedi**: Writing - review & editing.

Disclosure and Competing Interests Statement

The authors declare that they have no competing interests.

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Reproducibility Statement

This paper is fully self-contained for independent replication. No external code repository, proprietary data, or unpublished component is required.

Architecture. Flexi-JEGNN is specified completely in Section ???. The atom feature vector is 18-dimensional (12-element atom-type one-hot, normalised degree, formal charge, implicit hydrogen count, aromaticity, ring, chirality flags), computed from SMILES via RDKit [?] with no 3D input. The five distance levels (L0–L4) are defined unambiguously in Section ???.2, including the exact constant value for L0 (1.4 Å), the Dijkstra path formula for L1, the RDKit Distance Geometry midpoint for L2 (Gasteiger et al. [29]), the ETKDGv3+MMFF94 pipeline for L3 [11, 13], and the uniform [1.0, 8.0] Å draw for L4. Gaussian smearing uses 16 RBF centres evenly spaced in [0.0, 5.0] Å with threshold $\tau = 5.0$ Å (Eq. 1). The joint-edge feature $\mathbf{E}_{ij}$ is the 21-dimensional concatenation of the 16-dimensional RBF output and a 5-dimensional one-hot bond-type vector (Eqs. 2–3). Five JointEdgeConv layers with hidden dimension $d = 256$ and a PNA multi-aggregation readout (Eq. 4) produce a 768-dimensional graph embedding fed to a 2-layer MLP classifier (≈ 4.2 M parameters total).

Training protocol. All hyperparameters are fixed identically across every distance level, dataset, and model family (Section ???.3): AdamW optimiser, learning rate 10^{-4} , weight decay 10^{-4} , cosine annealing schedule, 80 epochs maximum, early stopping with patience 15, dropout $p = 0.3$, Layer Normalisation on every residual connection, positive-weighted Binary Cross-Entropy for classification, Pearson r as the regression metric for PDBbind. No per-condition hyperparameter search was performed; all values were fixed before any experimental run.

Datasets. All six benchmarks are publicly available: BACE, BBBP, HIV via MoleculeNet [35]; PDBbind [36]; QM9 [37]; TDC ADMET [38]. Murcko scaffold splits (80/10/10) [39] with $n=20$ independent random seeds reproduce the exact experimental conditions reported in Table ??.

Software. The implementation uses PyTorch Geometric 2.5.3 [41] (PyTorch 2.2.1, CUDA 12.1) for message-passing layers and RDKit 2023.09.5 [?] for all molecular featurisation and distance computation. Statistical tests use SciPy 1.12 [42]. All packages are open-source; explicit version numbers are given here and in Section ??3. To ensure full reproducibility, a Docker container specification (**Dockerfile**) and Conda environment (**environment.yml**) are provided in the supplementary code repository at <https://github.com/Anonymous/Flexi-JEGNN>.

Compute. Experiments were run on an NVIDIA RTX 6000 Ada (primary, 48 GB VRAM) and an NVIDIA RTX 4060 (secondary, 8 GB VRAM) under Windows 11. Total wall-clock time for all 3,250 completed runs was approximately 420 GPU-hours. Each individual run (one seed, one level, one dataset, one model) completes in under 15 minutes on the primary GPU.

A ANOVA Summary

[Table 7 here]

B Ablation: Layers and Hidden Dimension

To confirm that the insensitivity finding is not an artefact of the specific architectural choices, the authors ablated layer count ($\{3, 5, 7\}$) and hidden dimension ($\{128, 256, 512\}$) on BACE and QM9 at L0 and L3, using $n=5$ seeds. In all nine configurations, the L0–L3 restricted ANOVA remained non-significant ($p > 0.10$), and the absolute L0 vs. L3 performance difference did not exceed 0.015 AUC. These results confirm that the choice of 5 layers and hidden dimension 256 does not determine the paper’s conclusion.

C Threshold Sensitivity Analysis (τ)

The authors varied the edge-creation threshold $\tau \in \{3.0, 5.0, 8.0\}$ Å on BACE and QM9 ($n=5$ seeds, L0 and L3). At $\tau = 3.0$ Å, the number of edges drops substantially (mean 181 vs. 238 at the default $\tau = 5.0$ Å), reducing the model’s receptive field; at $\tau = 8.0$ Å, the graph becomes nearly complete for smaller molecules. In both cases, the restricted ANOVA (L0–L3) remained non-significant, confirming that the threshold choice does not determine the paper’s central conclusion.

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